

Ex.no:6b

DATA VISUALIZATION USING POWER BI – RETAILS_SALES_RAW

Aim:

To Load the data into PowerBI and transform the data and visualize it to get insights

Algorithm:

Algorithm: Sales Dashboard Calculation

Step 1: Input Data Loading

- Read the dataset from retail_sales_raw containing columns Product, Region, and Sales.

Dataset rows:

Laptop, North, 50000
Mobile, South, 35000
Printer, East, 25000
Laptop, West, 45000
Mobile, North, 40000
Printer, South, 30000

Step 2: Region Filter Check

Check slicer selection for Region.

Default selected regions: East, North, South, West.

Retain all rows matching the selected regions in Filtered Data.

Step 3: KPI Total Sales Calculation

Sum all values in the Sales column from Filtered Data.

Total Sales = 50000 + 35000 + 25000 + 45000 + 40000 + 30000 = 225000 (displayed as 225K).

Step 4: Bar Chart Calculation (Sum of Sales by Product)

Group Filtered Data by Product and sum Sales for each product:

- Laptop Sales = 50000 (North) + 45000 (West) = 95000
- Mobile Sales = 35000 (South) + 40000 (North) = 75000

- $\text{Printer Sales} = 25000 (\text{East}) + 30000 (\text{South}) = 55000$

Sort products in descending order by total sales: Laptop, Mobile, Printer.

Step 5: Donut Chart Calculation (Sum of Sales by Region)

Group Filtered_Data by Region and sum Sales for each region:

- $\text{North Sales} = 50000 + 40000 = 90000$
- $\text{South Sales} = 35000 + 30000 = 65000$
- $\text{West Sales} = 45000$
- $\text{East Sales} = 25000$

Calculate percentage share for each region relative to Total Sales (225000):

- $\text{North Share} = (90000 / 225000) * 100 = 40.00\%$
- $\text{South Share} = (65000 / 225000) * 100 = 28.89\%$
- $\text{West Share} = (45000 / 225000) * 100 = 20.00\%$
- $\text{East Share} = (25000 / 225000) * 100 = 11.11\%$

Step 6: Dashboard Render

Render KPI card with formatted total value 225K.

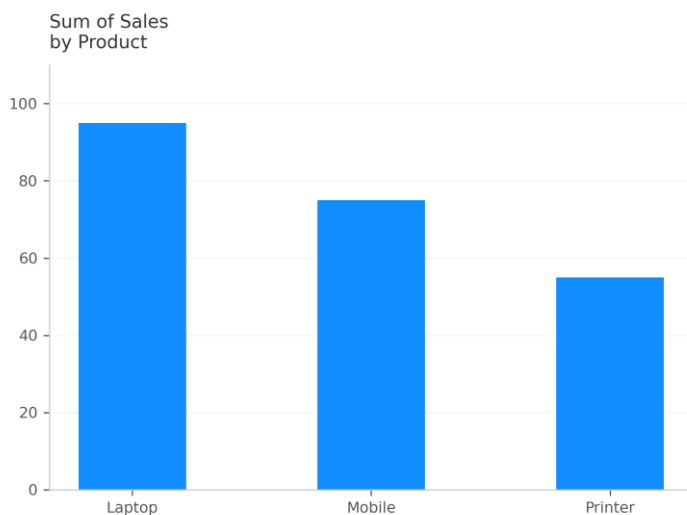
Render vertical bar chart with Laptop at 95K, Mobile at 75K, Printer at 55K.

Render donut chart with region segments North (90K, 40%), South (65K, 28.89%), West (45K, 20%), East (25K, 11.11%), and center value 225K.

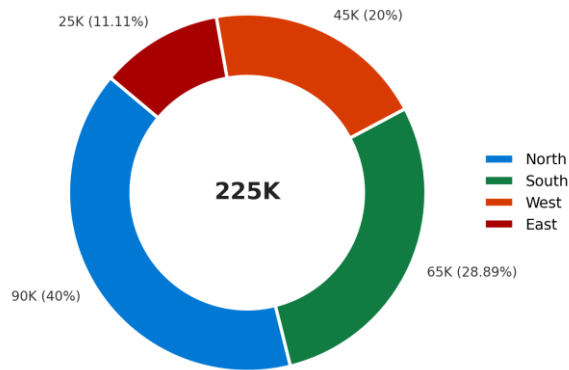
Render region slicer checkboxes for East, North, South, West.

On slicer selection update, filter rows and recompute steps 2 through 6.

Output:



Sum of Sales
by Region



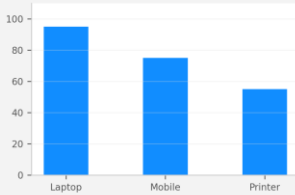
Sum of Sales

225K

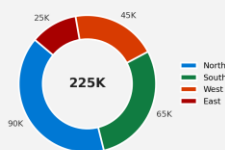
Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Sum of Sales by Product



Sum of Sales by Region



Sum of Sales

225K

Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Result:

Data visualization using Power BI with retail_sales_raw dataset was successfully executed and dashboard insights were generated.